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Project Portfolio

This document provides a detailed overview of selected academic, research, and software development projects.

Research Projects

Semantic Drift Analysis in Bengali (Undergraduate Thesis)

Technologies: Python, PyTorch, Transformers, OCR, NLP

- Conducting research on semantic drift in Bengali using diachronic corpora spanning over 120 years.
- Built OCR and preprocessing pipelines to curate a clean historical text corpus.
- Fine-tuned Historical BanglaBERT for contextual semantic analysis.
- Investigated changes in word meanings over time using transformer-based embeddings.

Leaf Disease Classification using TinyResNet

Technologies: Python, PyTorch, Computer Vision

- Designed a lightweight CNN model for tomato leaf disease classification.
- Proposed distortion-aware augmentation including blur, noise, and low-light conditions.
- Achieved 96.59% classification accuracy.
- Outperformed baseline ResNet18 in generalization under real-world distortions.

GitHub: [leaf-disease-research](#)

Real-Time Fraud Detection System

Technologies: Python, LightGBM, Scikit-learn

- Developed an end-to-end fraud detection pipeline for highly imbalanced credit card transaction data.
- Applied cost-sensitive learning and threshold tuning.
- Built a live transaction simulation console for real-time alert generation.
- Performed extensive EDA and model evaluation.

GitHub: [Fraud-Detection-Realtime](#)

Natural Language Processing Project

Token-Level Part-of-Speech Tagging using RNN Models

Technologies: Python, TensorFlow/Keras, RNN, LSTM, GRU, BiLSTM, NLP

- Developed a sequence labeling system for Part-of-Speech tagging using the POS 1 dataset containing over 19,000 training sentences and 41 unique POS tags.
- Implemented and compared Simple RNN, LSTM, GRU, and BiLSTM architectures with 300-dimensional word embeddings.
- Performed preprocessing including tokenization, label encoding, mismatch filtering, and sequence padding for token-level classification.
- Achieved the best performance with **BiLSTM**, reaching **97% accuracy** and a **0.85 macro F1-score**.

GitHub: [Repository Link](#)

Machine Learning Project

Disease Prediction using Machine Learning

Technologies: Python, Scikit-learn, Pandas, NumPy

- Developed a machine learning system to predict diabetes using the Pima Indians Diabetes Dataset containing 768 patient records and 8 clinical features.
- Applied data preprocessing techniques including mean imputation for missing values and feature scaling using **MinMaxScaler**.
- Implemented and compared **K-Nearest Neighbors**, **Decision Tree**, and **Naive Bayes** classifiers.
- Identified **Naive Bayes** as the best-performing model based on accuracy, precision, and F1-score.

GitHub: [Repository Link](#)

Software Development Project

Shikkha.com

Technologies: React, Vite, Node.js, Express.js, MongoDB, Mongoose, JWT, Stripe, Tailwind CSS, Framer Motion

- Developed a full-stack EdTech platform supporting course enrollment, book purchases, educator appointment scheduling, online exams, and automated grading.
- Built secure REST APIs with JWT authentication, role-based access control, file uploads, and MongoDB integration.
- Integrated **Stripe Checkout** for unified payments across courses and books, with webhook-based enrollment and purchase confirmation.
- Implemented AI-assisted features using the **Mistral API** to enhance learning and platform functionality.

- Designed a responsive frontend using React, Tailwind CSS, and Framer Motion to provide a smooth user experience.

GitHub: [Repository Link](#)

Hardware Project

Automated Sanitizer and Mask Dispensing Robot

Technologies: Arduino Uno, PIR Sensor, Ultrasonic Sensor, IR Sensors, DC Motor, Sanitizer Pump, 16×2 LCD

- Developed a low-cost, fully contactless hygiene system that automatically dispenses hand sanitizer and face masks upon detecting a user.
- Integrated PIR, ultrasonic, and IR sensors with microcontroller-based control logic to enable reliable human detection and smart trigger mechanisms.
- Implemented automated sanitizer pumping, motor-driven mask dispensing, and LCD-based user guidance for seamless interaction.
- Designed as a scalable solution for hospitals, educational institutions, and public facilities to promote hygiene and reduce physical contact.

GitHub: [Repository Link](#)

Networking Project

Federal Flight Agency Mission-Critical Aerospace Network Design

Technologies: Cisco Packet Tracer, RIPv2, Static Routing, VLSM, DNS, HTTP, SMTP/POP3

- Designed a high-availability network architecture for an autonomous aircraft fleet, integrating AI navigation systems, command centers, and emergency override infrastructure.
- Implemented efficient IP addressing using **VLSM** and configured a hybrid routing strategy combining RIPv2, static routes, and floating static routes for automatic failover.
- Applied security measures including passive interfaces, route filtering, and isolation of critical emergency nodes from dynamic routing protocols.
- Deployed centralized network services including DNS, web, and email servers to support mission-critical communication and operations.

GitHub: [Repository Link](#)